SSIE 605 | Binghamton University

Using Logistic Regression to Determine U.S. Equity Performance

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Q1 2010 – Q4 2024 | S&P 500 Constituents

PREDICTING OUTPERFORMANCE USING QUARTERLY FINANCIAL RATIOS

Logistic Regression | VIF | PCA | Out-of-Sample 2025

Agenda

Presentation Structure

Background

Section 01

The SEC requires all publicly traded companies to disclose quarterly financial statements. Thes
e reports detail revenues, expenses, assets, and debt — information used to evaluate company health.

S&P 500

500 largest U.S. companies by market cap.

Covers ~80% of total U.S. stock market value.

Rebalanced quarterly as constituents change.

Passive Investing Assumption

The average investor cannot consistently beat the S&P 500 return.

Index funds: low fees, no stock selection needed.

Our Research Question

Can publicly available quarterly financial data predict which S&P 500 companies will outperfo
the index the following quarter?

Problem Statement

Section 02

Q1

Predictability

Can we use publicly available S&P 500 financial data to determine if a company will outperform or underperform the S&P 500?

Q2

Forward-Looking Signal

Can we use current quarter financial ratios to reliably predict next quarter stock performance — before it happens?

Q3

Portfolio Construction

Can we construct a portfolio from model predictions whose quarterly returns beat the S&P 500 index?

Data Capture — Sources

Section 03

WRDS

Wharton Research Data Service

University of Pennsylvania data repository.

Source: S&P 500 constituent history,

quarterly financials (Compustat),

and stock returns (CRSP).

Coverage: Q1 2010 – Q4 2024.

FRED

Federal Reserve Economic Data

Federal Reserve Bank of St. Louis.

Source: GDP growth and CPI inflation.

One macro value per quarter,

matched to each company's feature quarter.

Accessed via FRED public API.

EDGAR

Electronic Data Gathering

SEC's official company filing repository.

Supplement for 10-Q financial data.

Accessed via API through Python.

Provides additional quarterly fields

not covered by Compustat alone.

All data accessed via API and processed with Python (pandas, statsmodels, scikit-learn)

Data Capture — Raw Variables

Section 03 — Compustat Quarterly Fields

prccq	Price Close (Quarter End)	**cshoq**	Common Shares Outstanding	**mkvaltq**	Market Capitalization	**revtq**	Total Revenue
cogsq	Cost of Goods Sold	**xsgaq**	SG&A Expenses	**xrdq**	R&D Expenses	**oiadpq**	Operating Income After Depr.
niq	Net Income	**ibq**	Income Before Extraordinary Iter	**piq**	Pretax Income	**atq**	Total Assets
ceqq	Common Equity Total	**teqq**	Total Stockholders Equity	**dlttq**	Long-Term Debt	**dlcq**	Debt in Current Liabilities
actq	Current Assets	**lctq**	Current Liabilities	**cheq**	Cash & Short-Term Investments	**dpq**	Depreciation & Amortization
oibdpq	Operating Income Before Depr.						

Source: WRDS Compustat Fundamentals Quarterly (fundq table) | Joined on gvkey identifier

Financial Ratios — 14 Base Ratios

Section 04 — Derived from Raw Variables

Return on Assets (ROA)	Net Income / Total Assets	Profitability
Return on Equity (ROE)	Net Income / Common Equity	Profitability
Gross Margin	(Revenue - COGS) / Revenue	Profitability
Operating Margin	Op. Income / Revenue	Profitability
Net Margin	Net Income / Revenue	Profitability
Asset Turnover	Revenue / Total Assets	Efficiency
Current Ratio	Current Assets / Current Liabilities	Liquidity
Debt to Equity	Long-Term Debt / Common Equity	Leverage
Revenue Growth	(Rev - Lag Rev) / |Lag Rev|	Growth
Net Income Growth	(NI - Lag NI) / |Lag NI|	Growth
P/E Ratio	Price / (EPS)	Valuation
Book-to-Market	Common Equity / Market Cap	Valuation
GDP Growth	Quarter-over-quarter GDP change (FRED)	Macro
Inflation	Quarter-over-quarter CPI change (FRED)	Macro

Lag values computed via groupby(gvkey).shift(1) | GDP and Inflation matched per quarter

Delta Features — Quarter-over-Quarter Momentum

Section 04 — 10 Additional Features

Delta features capture the direction of change in each ratio — whether fundamentals are impro
ving or deteriorating. Two companies with the same ROA tell very different stories if one is rising and the other falling.

Formula & Example

Delta_Ratio(Q) = Ratio(Q) - Ratio(Q-1)

Computed via groupby(gvkey).diff()

Example:

ROA (Q3 2024) = 5.0%

ROA (Q4 2024) = 8.0%

Delta_ROA = +3.0% (Improving)

Positive delta = improving fundamentals

Negative delta = deteriorating fundamentals

10 Delta Features Added

- Delta ROA
- Delta ROE
- Delta Gross Margin
- Delta Operating Margin
- Delta Net Margin
- Delta Asset Turnover
- Delta Current Ratio
- Delta Debt/Equity
- Delta P/E Ratio
- Delta Book-to-Market

Total features entering VIF analysis: 24 (14 base ratios + 10 delta features)

Outperformer Definition & Forward-Looking Label

Section 04

Label Construction — Forward-Looking by One Quarter

Stock Return (Q+1) > S&P 500 Return (Q+1) --> Label = 1 (Outperformer)

Stock Return (Q+1) <= S&P 500 Return (Q+1) --> Label = 0 (Underperformer)

Feature Quarter: Q4 2024

Financial ratios from 10-Q filings.

14 base ratios + 10 delta features.

Available before Q1 2025 begins.

Features are derived from company financials that are public record by the time of prediction.

-->

Label Quarter: Q1 2025

Was stock's Q1 2025 return greater than SPY Q1 2025 return?

Yes -> Label = 1 (Outperformer)

No -> Label = 0 (Underperformer)

Computed via CRSP quarterly returns.

`outperformer_next = groupby(gvkey)[outperformer_quarterly].shift(-1)` | Training data: 16,589 rows

Data Cleaning

Section 05 — Winsorization, Standardization, NaN Removal

Step 1: Winsorization

Extreme outlier values capped at 1st and 99th percentile

Example: If ROA = -300%, it is capped at -15%
(the 1st percentile value). The row is kept;
only the extreme value is replaced.

Applied per ratio using training data bounds.
Test data uses same bounds to prevent data leakage.

Step 2: Standardization (Z-Score)

Each ratio scaled to mean=0, std=1.

Without this, P/E Ratio (50x) would dominate
over ROA (0.05), biasing the model.

Ensures all 24 features on the same scale
before VIF analysis and PCA.

Step 3: NaN Removal

Rows with any missing values are dropped.

Common causes:

- Companies with zero revenue (div by zero)
- First entries (no lag for delta features)

Result: 21,285 -> 16,589 complete rows.
(-22% of observations removed)

Rows Before
21,285
training observations

->>

Rows After
16,589
complete rows

VIF Analysis — Multicollinearity Removal

Section 06 — Variance Inflation Factor, Threshold = 2.5

Financial ratios are correlated — e.g., Net Margin and ROA both measure profitability. VIF detects this collinearity. Features with VIF > 2.5 are iteratively removed.

VIF Procedure

1. Compute VIF for all 24 features
2. Find the feature with highest VIF
3. If VIF > 2.5, remove that feature
4. Repeat until all VIF <= 2.5
5. Threshold of 2.5 is stricter than the standard 5.0 cutoff (Ananthakumar & Sarkar 2017)

20 Features Retained (of 24)

roe	gross_margin
op_margin	asset_turnover
current_ratio	debt_to_equity
rev_growth	ni_growth
pe_ratio	book_to_market
gdp_growth	inflation
delta_roe	delta_gross_margin
delta_net_margin	delta_asset_turnover
delta_current_ratio	delta_debt_to_equity
delta_pe_ratio	delta_book_to_market

Retained
20/24
VIF <= 2.5

Removed
4
VIF > 2.5

Threshold
2.5
Paper benchmark

PCA Analysis — Dimensionality Reduction

Section 07 — Principal Component Analysis, Threshold >= 5%

Even after VIF filtering, 20 correlated features remain. PCA compresses them into uncorrelated components, each capturing a distinct pattern — eliminating residual multicollinearity.

Component Selection Rule

Threshold: >= 5% individual variance explained

Applied on faculty advisor recommendation.

Rule: n_comp = max(1, count of
components where var. ratio >= 0.05)

Result: 9 principal components retained
for the global model.

Components vary per sector/configuration.

Component Themes (Global Model)

PC1	Profitability cluster — ROA, margins
PC2	Leverage & liquidity — Debt/Equity, Current Ratio
PC3	Revenue momentum — Rev Growth, Delta Rev
PC4	Valuation — P/E Ratio, Book-to-Market
PC5	NI & margin delta — Delta Net Margin, Delta ROE
PC6	Asset efficiency — Asset Turnover
PC7	Macro environment — GDP Growth, Inflation
PC8	Equity momentum — Delta ROE, Delta B/M
PC9	Mixed residual signal

Components Kept
9
Global model

Threshold
>=5%
Per component

Significant
4/9
$p<0.05$: PC1,PC3,PC6,PC7

Logistic Regression — Model Setup

Section 08 — Global Pooled Model

With PCA components as inputs, logistic regression models the probability that a stock will o
utperform the S&P 500 the following quarter. Trained on Q1 2010 – Q4 2024.

Training Period	**Q1 2010 – Q4 2024**	15 years of quarterly data
Training Rows	**16,589**	complete rows after cleaning
Companies	**~500**	S&P 500 constituents (time-varying)
Target Variable	**outperformer_next**	1 = beats SPY next Q, 0 = does not
Input Features	**9 PCA Components**	derived from 20 VIF-filtered ratios
Model	**Logistic Regression**	statsmodels Logit, max_iter=200
Cutoff Optimization	**Train-set threshold**	optimized over 0.30 – 0.75 range
Class Weights	**Balanced**	accounts for outperformer imbalance

Benchmark: Ananthakumar & Sarkar (2017) — 71.2% accuracy on S&P 500 using logistic regression

Global Model Results

Section 08 — All S&P 500, No Segmentation

Test Accuracy	LR AUC	GB AUC	Sig. PCA Components
53.2%	**0.5381**	**0.6589**	**4 / 9**
Global pooled model	Logistic Regression	Gradient Boosting benchmark	$p < 0.05$

Key Interpretation

AUC 0.5381:	Near-random discrimination globally. The model finds weak but real signal.
53.2% Accuracy:	Marginally above 50% baseline. Markets are competitive to predict.
Statistically Sig.:	p-value < 0.05 — model is not random; it finds true signal.
GB AUC 0.6589:	Non-linear relationships exist that LR cannot fully capture.
McFadden R2:	0.0073 — weak but significant.

Performance vs. Benchmark

Metric	Our Result	Benchmark (2017)
Accuracy	53.2%	71.2%
AUC	0.5381	~0.75 (est.)
Sensitivity	~52%	Not reported
Specificity	~54%	Not reported
Segmentation	Best: 63.9%	Single model

Gradient Boosting — Non-Linear Benchmark

Section 08 — LR vs GB Comparison

To test the linearity assumption, Gradient Boosting (GB) was run as a diagnostic benchmark. GB captures non-linear patterns that logistic regression cannot model.

Global Model Comparison

Model	Train AUC	Test AUC	Role
Logistic Regression	0.5381	0.5381	Primary model — interpreta
Gradient Boosting	0.6589	0.6589	Diagnostic benchmark only

AUC Improvement (GB vs LR): +12.1 percentage points

What This Means

The gap: LR (0.54) vs GB (0.66) confirms non-linear patterns exist.

LR finds real, significant signal but misses interaction effects between ratios.

GB is used ONLY as a diagnostic tool.
LR remains the primary model because:

- Interpretable coefficients
- Odds ratios
- p-values for significance testing
- Required for academic analysis

Future work: explore GB, Random Forest, and neural network models.

Section 09

Segmentation & Results

Market Cap | Sector GICS | Clustering

Segmentation — Market Capitalization

Section 09 — Configuration 1

Companies segmented by market cap. Separate LR model trained and evaluated per size tier.

Large Cap

> $10B

Accuracy

54.2%

AUC: 0.558

Most institutional coverage.

Efficient pricing — harder to predict.

Mid Cap

$2B – $10B

Accuracy

52.8%

AUC: 0.542

Some analyst coverage.

More earnings variability.

Small Cap

< $2B

Accuracy

54.5%

AUC: 0.535

Limited analyst coverage.

Higher fundamental variability.

Conclusion: Market Cap alone provides only marginal improvement over global 53.2% baseline.

Segmentation — GICS Sector

Section 09 — Configuration 2

Separate LR model per GICS sector. Financial ratios mean different things across sectors — sector-specific models find stronger patterns.

Sector	N Obs	Accuracy	AUC
Communication Services	428	**59.5%**	0.6059
Consumer Discretionary	2668	54.1%	0.5305
Consumer Staples	1030	55.7%	0.5510
Energy	261	**61.7%**	0.6036
Financials	4108	55.0%	0.5350
Health Care	1126	54.3%	0.5578
Industrials	4183	54.5%	0.5350
Information Technology	2240	53.2%	0.5420
Materials	2781	53.2%	0.5481
Real Estate	424	**59.5%**	0.6010
Utilities	1757	54.5%	0.5420

Best: Communication Services (59.5%, AUC 0.606) | Energy (61.7%) | Real Estate (59.5%)

Clustering within Comm. Services improved further to 63.9% accuracy — best across all 45 configurations

Best Config — Communication Services + Clustering

Section 09 — Config 4: Sector + K-Means (k=2)

Within Communication Services, K-Means (k=2) split the sector into two sub-groups with distinct financial profiles. Cluster 0 (122 companies) was the best of all 45 configurations tested.

Best AUC	Best Accuracy	Weighted AUC	Configs Tested
0.6902	**63.9%**	**0.5720**	**45**
Comm. Services — Cluster 0	122 observations	Across all 45 configs	All sector x cluster combos

Top 4 Configurations (of 45)

1st	**Comm. Services — Cluster 0**	AUC 0.6902	**63.9%**
2nd	Comm. Services — Cluster 1	AUC 0.6099	58.0%
3rd	Communication Services	AUC 0.6059	59.5%
4th	Comm. Services + MktCap	AUC 0.6023	59.0%

Why Communication Services?

Companies like Netflix, Meta, Google, Disney have highly distinctive financial ratio patterns.

Subscriber growth, content investment, and advertising revenue create strong fundamental differentiation.

Clustering found a subscription-driven sub-group (Cluster 0) with especially strong predictive signal.

Benchmark: Ananthakumar & Sarkar 71.2%

Our best result: 63.9%

2025 Out-of-Sample Testing

Section 10 — True Forward-Looking Validation

The model trained on Q1 2010 – Q4 2024 was applied to predict S&P 500 outperformers for all fou
r quarters of 2025 — data the model never saw during training.

Q4 2024 ->

Q1 2025

334 stocks

Accuracy

44.6%

New addition — first true OOS

Q1 2025 ->

Q2 2025

403 stocks

Accuracy

55.3%

SPY +10.6% (strong market)

Q2 2025 ->

Q3 2025

409 stocks

Accuracy

56.2%

Largest 2025 sample

Q3 2025 ->

Q4 2025

73 stocks

Accuracy

57.5%

Partial: early filers only

Note: Q4 2025 has only 73 stocks — Q3 2025 financials were not fully filed at extraction time.
Early filers tend to be larger, well-governed companies. Results for this quarter should be interpreted cautiously.

2025 Portfolio vs S&P 500

Section 10 — $10,000 Simulated Portfolio

Portfolio constructed each quarter from predicted outperformers, allocated proportionally by ma rket cap. Compared against same $10,000 invested in SPY (S&P 500 ETF).

Quarter	Stocks	Pred. Out.	Model Return	SPY Return	Alpha	Beat SPY
Q2 2025	387	172	+6.39%	+10.57%	-4.18%	NO
Q3 2025	391	211	+6.74%	+7.79%	-1.06%	NO
Q4 2025	72	33	+3.15%	+2.35%	+0.81%	YES

Summary

Q2 & Q3 2025: Model underperformed SPY by -4.2% and -1.1%. Q4 2025: Beat SPY by +0.8% (partial, 72 stocks).

Overall: model did not consistently beat passive S&P 500 investing — consistent with efficient market theory.

Q1 2025: SPY returned -4.6% (bear market quarter); Model returned -2.2% — less negative, but accuracy was 44.6%.

Conclusion

Section 11 — Findings and Efficient Market Hypothesis Assessment

Finding 1: Logistic Regression Finds Weak Signal

Global model achieves 53.2% accuracy, AUC 0.5381 — statistically significant (p<0

Quarterly financial ratios contain predictive information; effect is real but modest.

Finding 2: Segmentation Improves Performance

Sector segmentation consistently outperforms global pooling.

Comm. Services + Clustering: 63.9% accuracy, AUC 0.6902 — best of all 45 config

Finding 3: Non-Linear Signal Exists

Gradient Boosting achieves AUC 0.6589 vs LR's 0.5381 globally.

Non-linear patterns (interaction effects) exist that LR cannot capture.

Future work: explore GB, Random Forest, and neural networks.

Finding 4: Portfolio Does Not Beat SPY Consistently

2025 portfolio underperformed in Q2 (-4.2%) and Q3 (-1.1%), beat narrowly in Q4 (-

Consistent with semi-strong form efficient market hypothesis.

EMH Verdict: We PARTIALLY SUPPORT the efficient market hypothesis.

Markets are hard to beat with public data, but are not perfectly efficient — sector-specific patterns yield significant predictive power.

Thank You

Questions welcome

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Key Statistics

Training Period	**Q1 2010 – Q4 2024**
Training Rows	**16,589 complete observations**
Features	**24 (14 base + 10 delta)**
VIF Filtered	**20 of 24 retained (threshold 2.5)**
PCA Components	**9 (global model, >=5% each)**
Global LR AUC	**0.5381**
Global LR Accuracy	**53.2%**
Best Accuracy	**63.9% — Comm. Services Cluster 0**
Best AUC	**0.6902 — Comm. Services Cluster 0**
2025 Predictions	**1,219 stocks across 4 quarters**
Benchmark (2017)	**Ananthakumar & Sarkar — 71.2%**